

# MATH 4162/Math 6201

## Numerical methods for differential equations

### Fall 2025

#### Homework problem set #1

Show all necessary steps of working in the given space

Name: \_\_\_\_\_

ID: \_\_\_\_\_

Upload the assignment solution no later than

23:59 on Sep 26, 2025

Full marks 50

1. (based on week 1)[15 pts]

(a) Consider the KdV equation on  $\mathbb{R}$

$$\frac{\partial u}{\partial t} - 6u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0$$

with decay conditions  $u, u_x, u_{xx} \rightarrow 0$  as  $|x| \rightarrow \infty$  for each fixed  $t$ . Show that the “mass”

$$M(t) = \int_{-\infty}^{\infty} u(x, t), dx$$

is constant in time.

(b) Use the Fourier transform pair given as

$$\hat{u}(k, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} u(x, t) e^{ikx} dx \quad \text{and} \quad u(x, t) = \int_{-\infty}^{\infty} \hat{u}(k, t) e^{-ikx} dk.$$

and consider

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^3 u}{\partial x^3}, \quad x \in \mathbb{R}, \quad t > 0,$$

with  $u(x, 0) = f(x)$  and  $u, \partial_x^m u \rightarrow 0$  as  $|x| \rightarrow \infty$ .

Convert the PDE into a system of ODE and finally, using inverse Fourier transform, show that

$$u(x, t) = \int_{-\infty}^{\infty} e^{(ik - k^2 + ik^3)t} \hat{f}(k) dk.$$

2. (based on week 2) [20 pts] The 1D wave equation is

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad x \in \mathbb{R}, \quad t \geq 0.$$

- (a) Derive the dispersion relation  $\omega = \omega(k)$  for the wave equation, compute the phase speed  $c_p$  and the group speed  $c_g$ , state whether the PDE is dispersive and/or dissipative, and justify your answer briefly.
- (b) Using Taylor series expansion about  $x_j$ , derive a five-point central finite difference approximation of  $\frac{\partial^2 u}{\partial x^2}$  using the five-point stencil  $\{x_{j-2}, x_{j-1}, x_j, x_{j+1}, x_{j+2}\}$  and report the lead error term.
- (c) Derive the dispersion relation of the wave equation discretized in space with three-point stencil and provide a plot by comparing the exact dispersion relation with the numerical dispersion relation.

3. [10pts] Consider the function

$$u(x) = e^{-\frac{x^2}{2\sigma^2}}, \quad x \in [-1, 1], \quad \sigma = 0.02.$$

- (a) Present a plot comparing exact  $\frac{\partial u}{\partial x}$  with its numerical approximation.
- (b) How many grid points are needed for 2-norm error to be less than  $10^{-2}$ ?